# Domain Adaptation for Acoustic Scene Classification

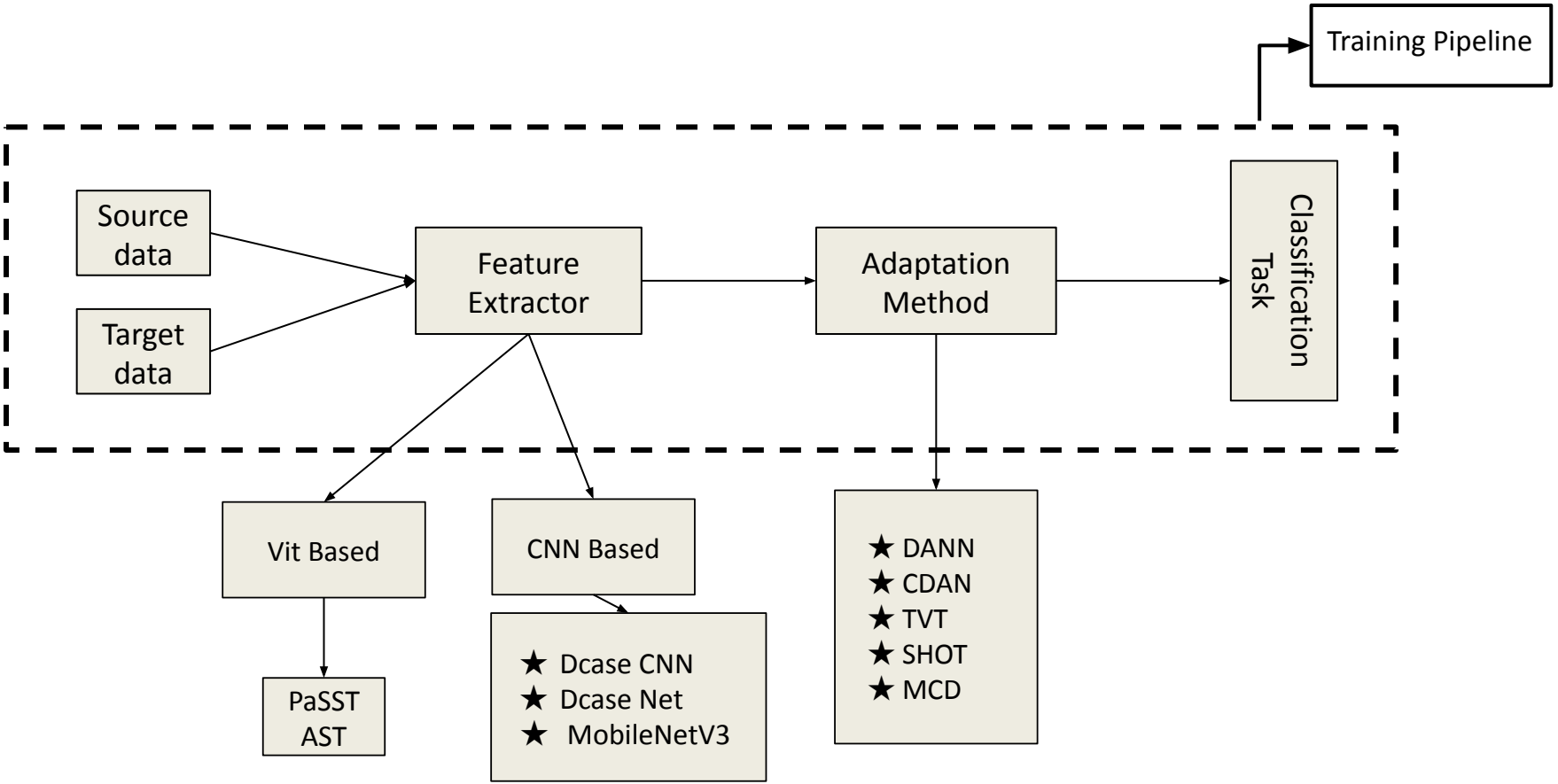
Abhishek Dileep

Indian Institute of Technology Mandi

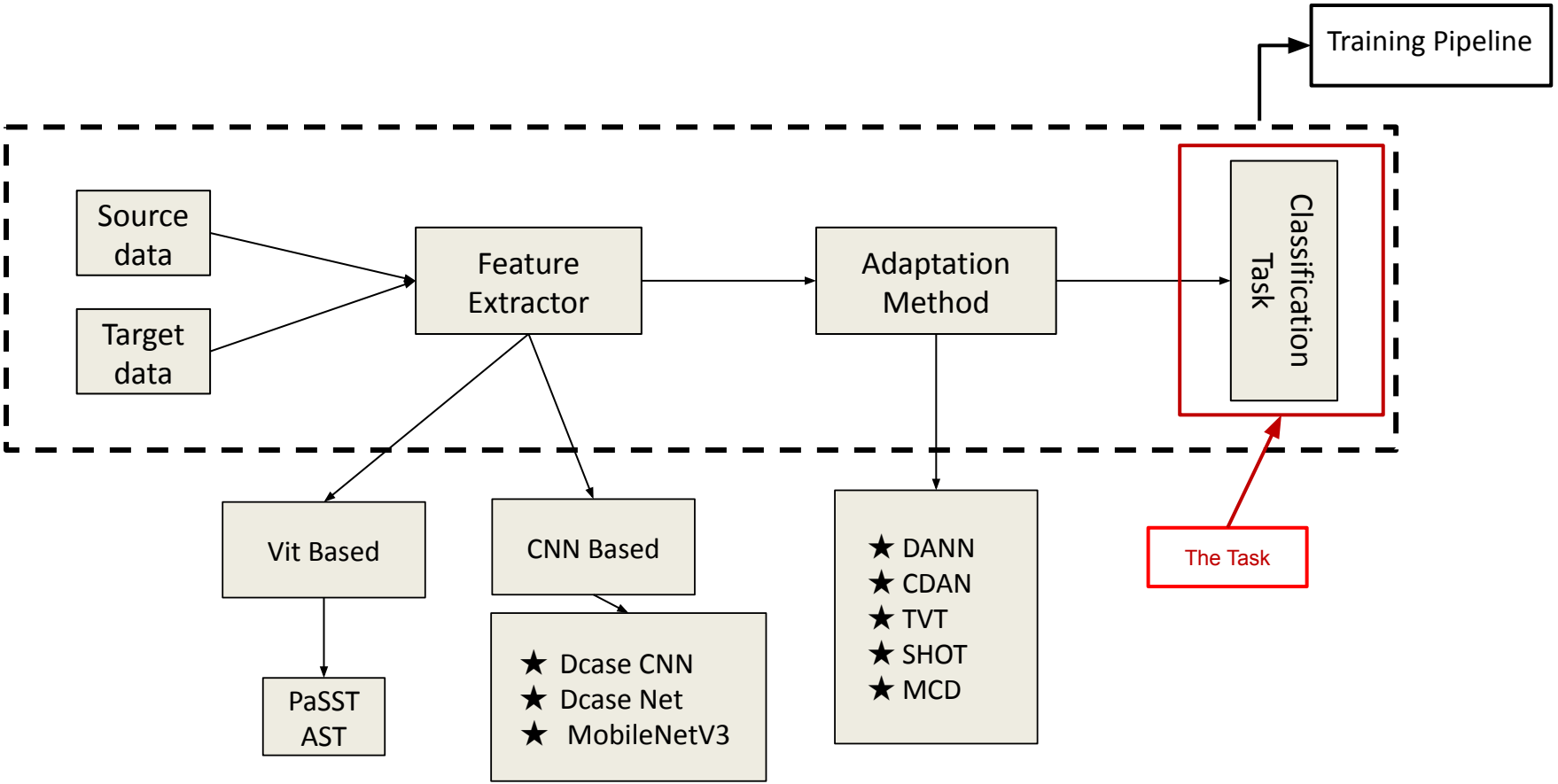
# Outline

- Overview
- Introduction
- Problem Formulation
- Domain Adaptation Methods
- Feature extractors
- Experimental Setup
- Results

# Overview


Training Pipeline
Source data
Target data
Feature Extractor
Adaptation Method
Classification Task
Vit Based
CNN Based
★ DANN
★ CDAN
★ TVT
★ SHOT
★ MCD
PaSST
AST
★ Dcase CNN
★ Dcase Net
★ MobileNetV3

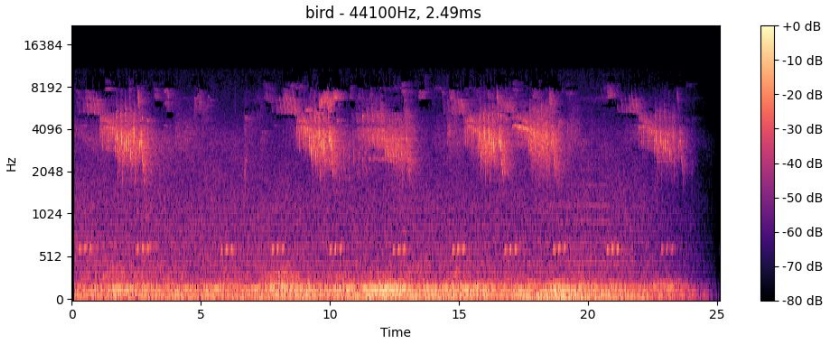
# Overview

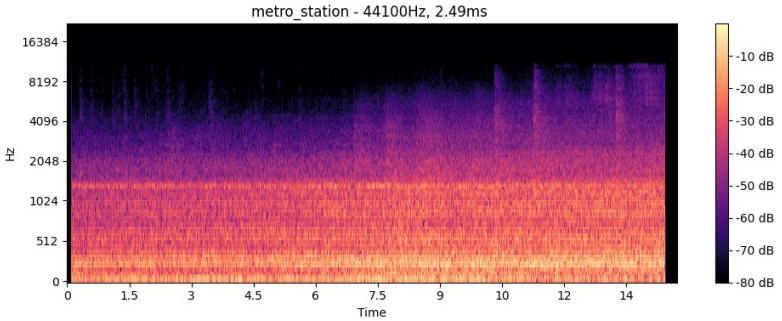

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# Introduction: Acoustic Scene Classification (ASC)



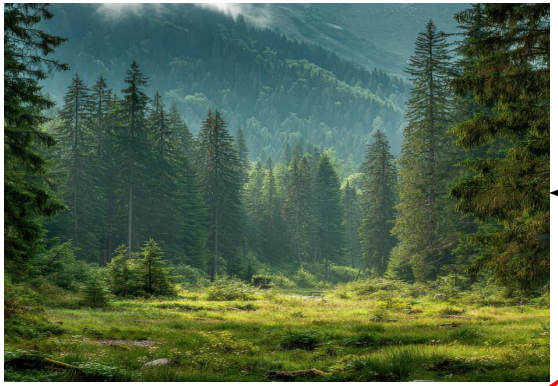




# Introduction: Acoustic Scene Classification (ASC)

Forest

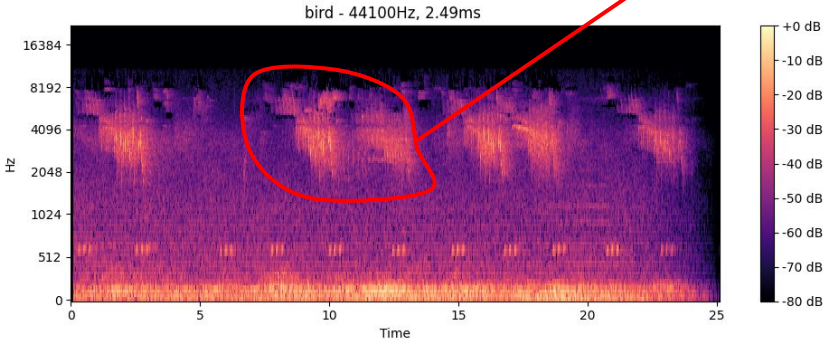


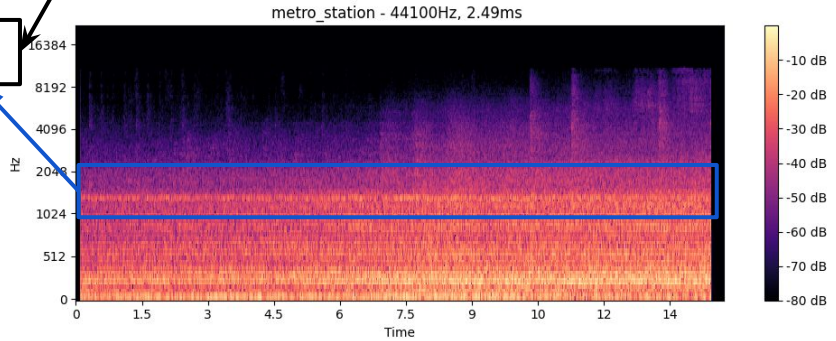
Metro station



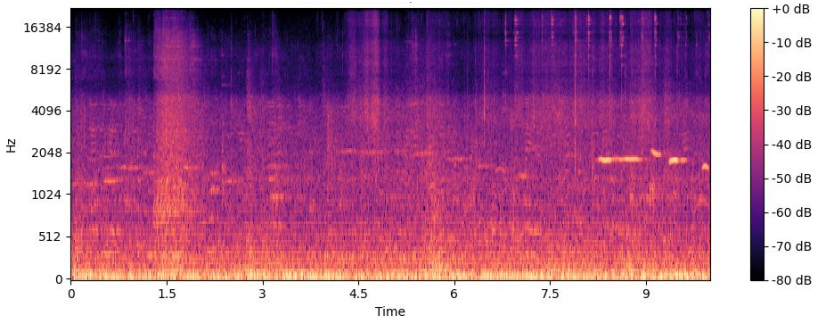
Multiple
Bird chirps

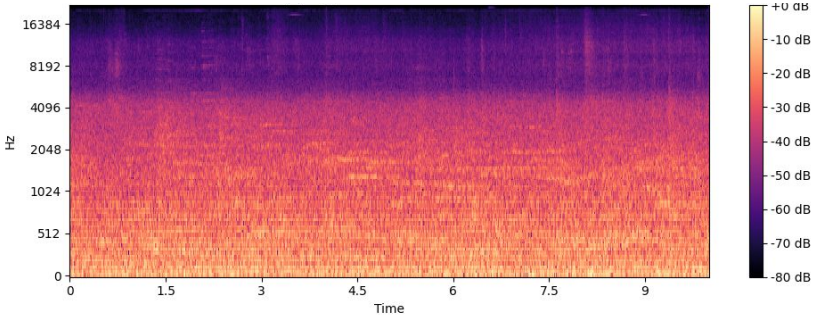
Metro



# Introduction: Acoustic Scene Classification (ASC)

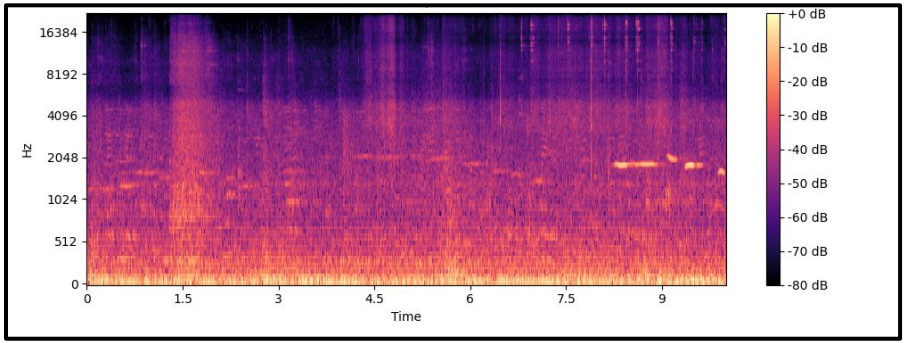


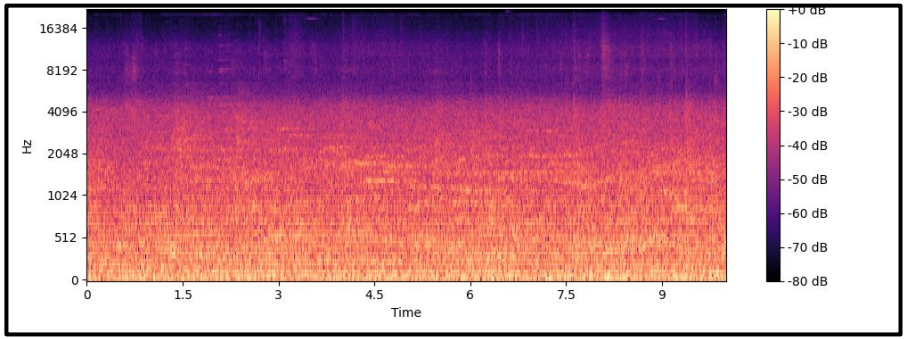








# Introduction: Acoustic Scene Classification (ASC)







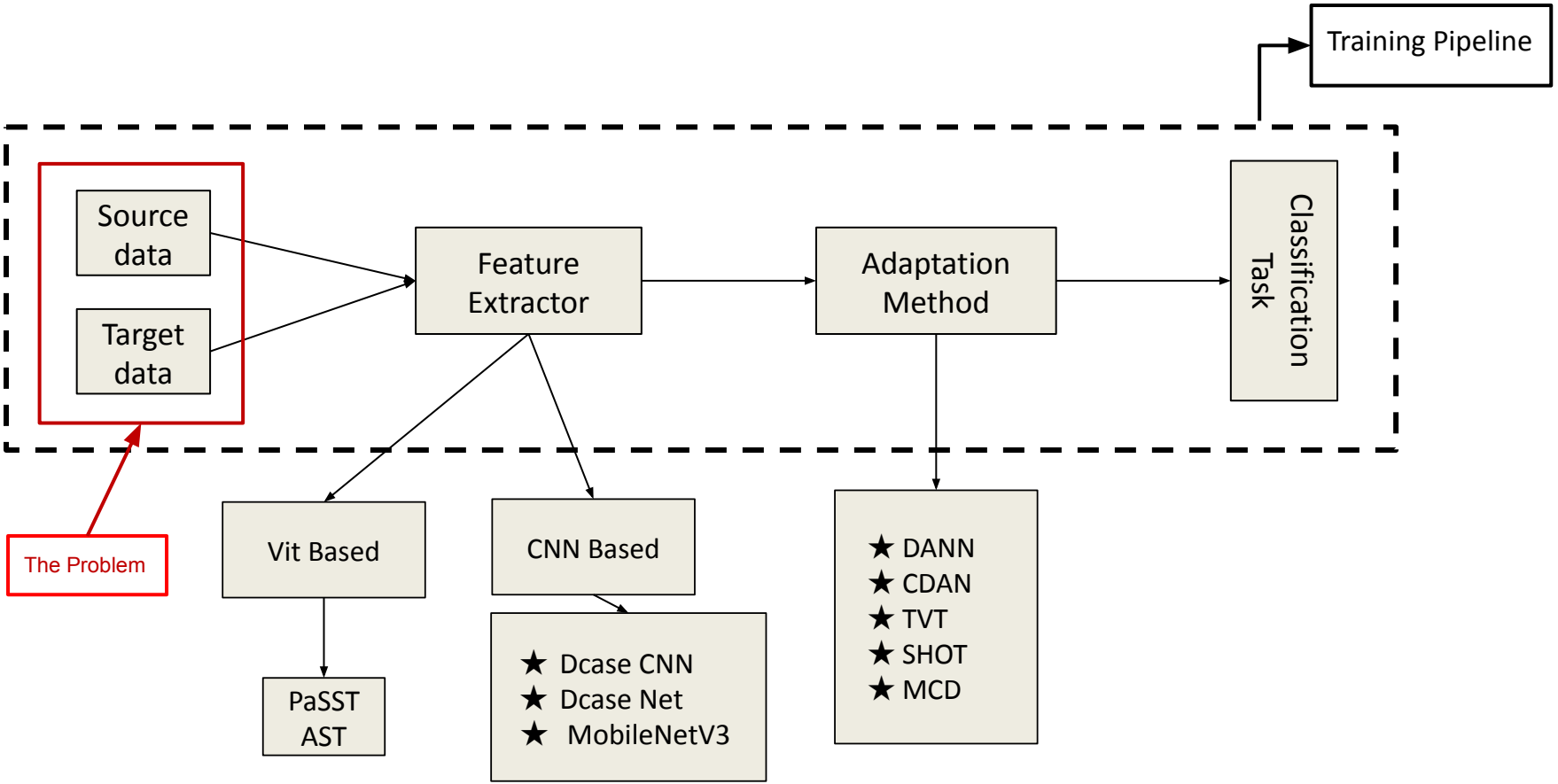

Airport

Perpetually Indistinguishable : Humans

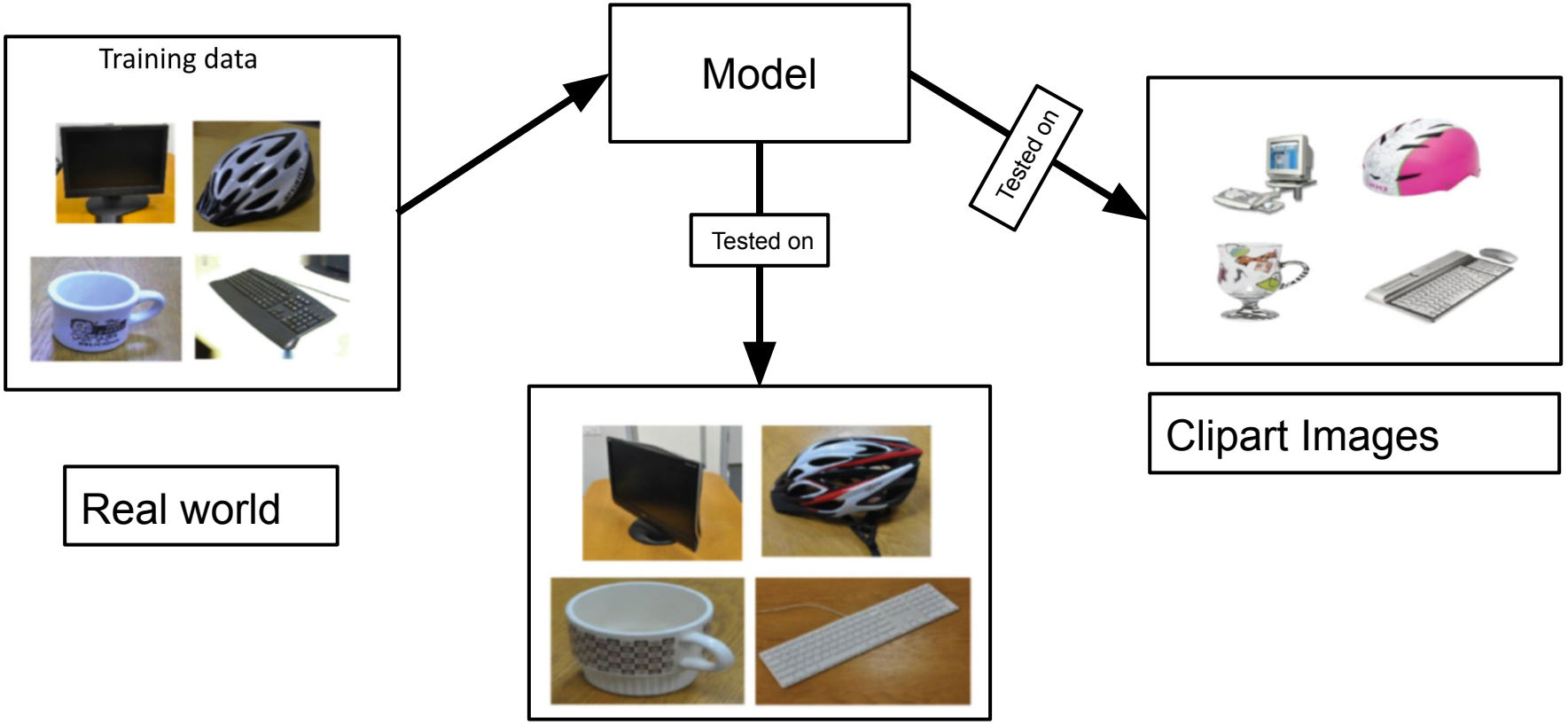
Shopping Mall

Barchiesi, Plumbey , et al. “Acoustic Scene Classification: Classifying Environments from the Sounds They Produce.” IEEE Signal Processing Magazine 2014,

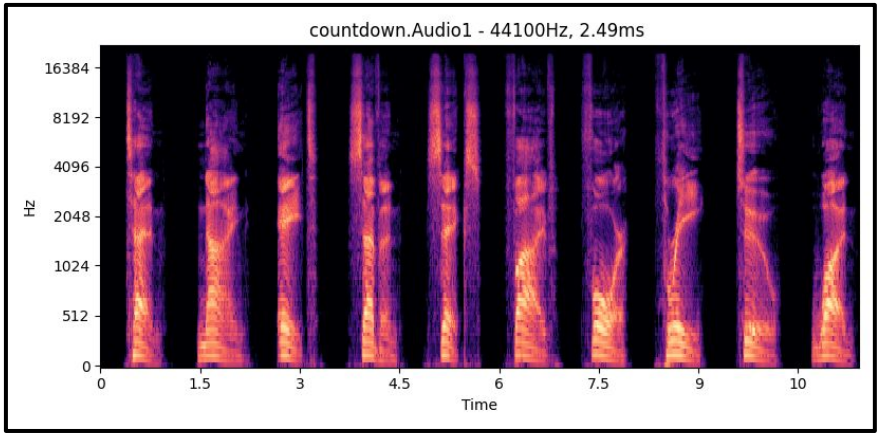
# Overview

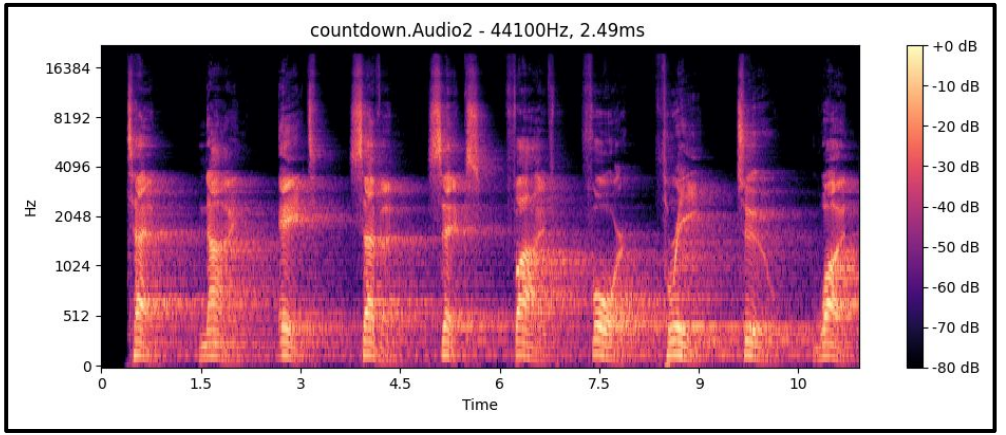

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# Introduction: Domain Shift



# Introduction: Domain Shift

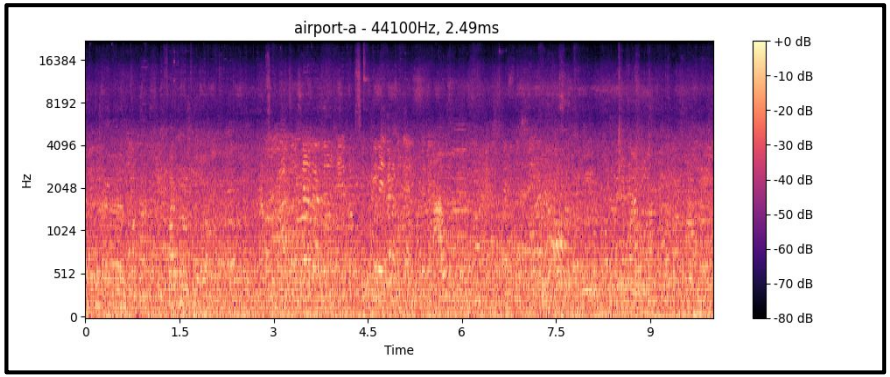


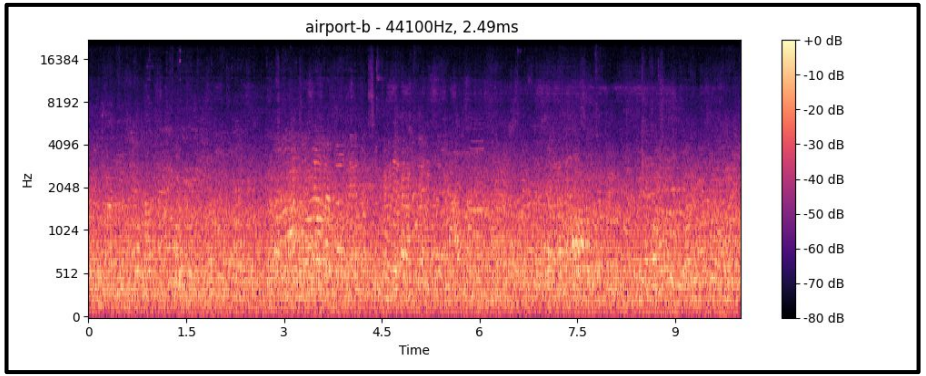




Countdown to 10

Reverb added

# Introduction: Domain Shift







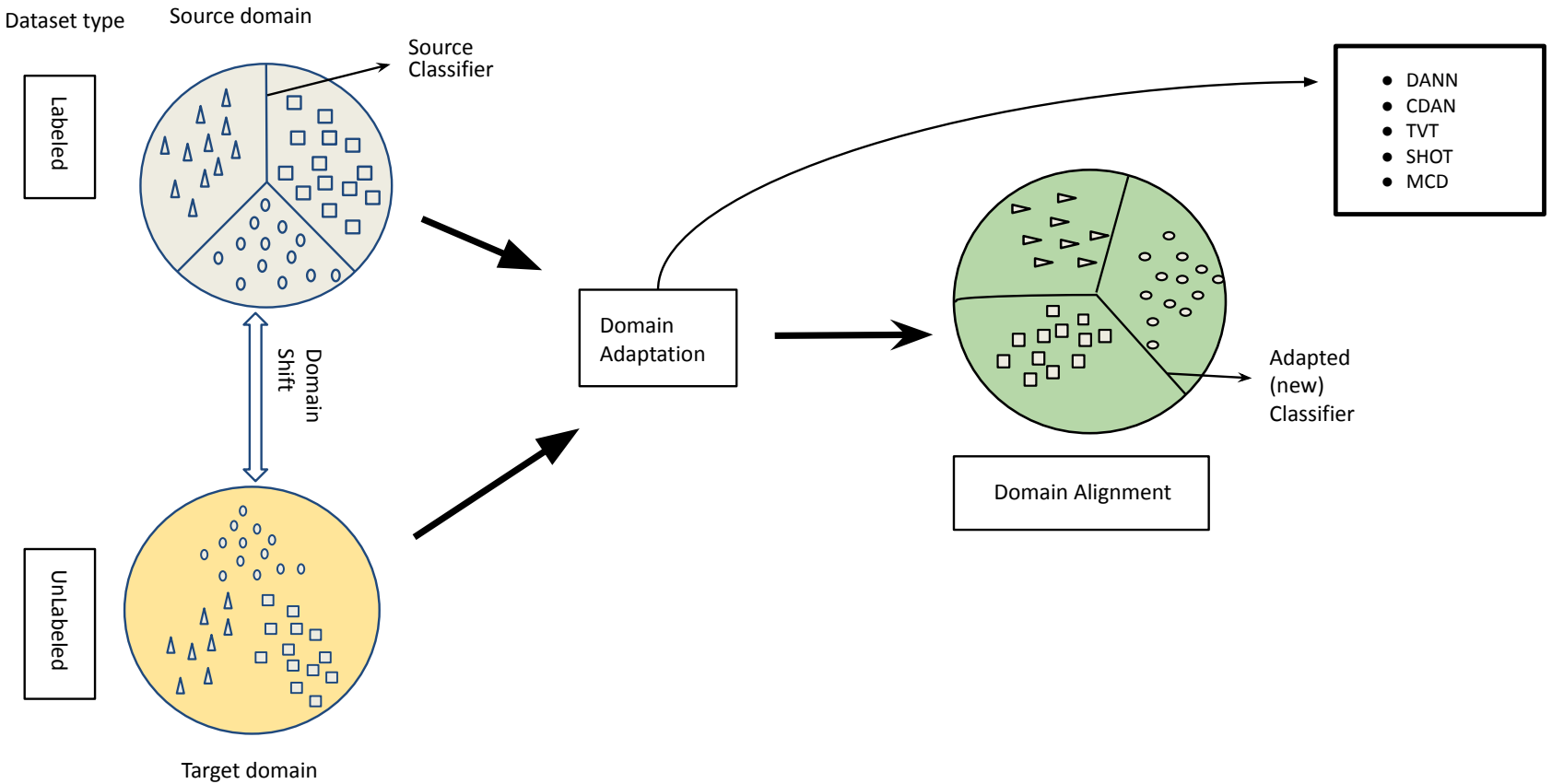





Device A : Soundman OKM II Klassik/studio A3 (Binaural Microphone)

Device B is a Samsung Galaxy S7

# Solution: Domain Adaptation



# Dataset : TAU 2020 & 2022 Dataset for Acoustic scene

## Class Labels

- Airport - `airport`
- Indoor shopping mall - `shopping_mall`
- Metro station - `metro_station`
- Pedestrian street - `street_pedestrian`
- Public square - `public_square`
- Street with medium level of traffic - `street_traffic`
- Travelling by a tram - `tram`
- Travelling by a bus - `bus`
- Travelling by an underground metro - `metro`
- Urban park - `park`

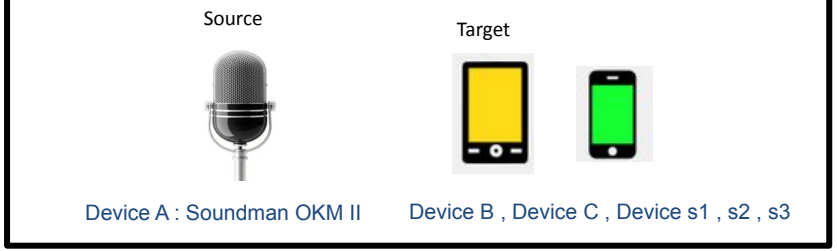
Main task : classification

- Variability between classes
- Variability between Devices

Sub task : Decrease Domain Shift

## Device Labels

- Device A : Soundman OKM II Klassik/studio A3 (Binaural Microphone)
- Device B is a Samsung Galaxy S7
- Device C is iPhone SE
- Device s1-s3 (simulated device from these original devices)



# Problem : Device Shift

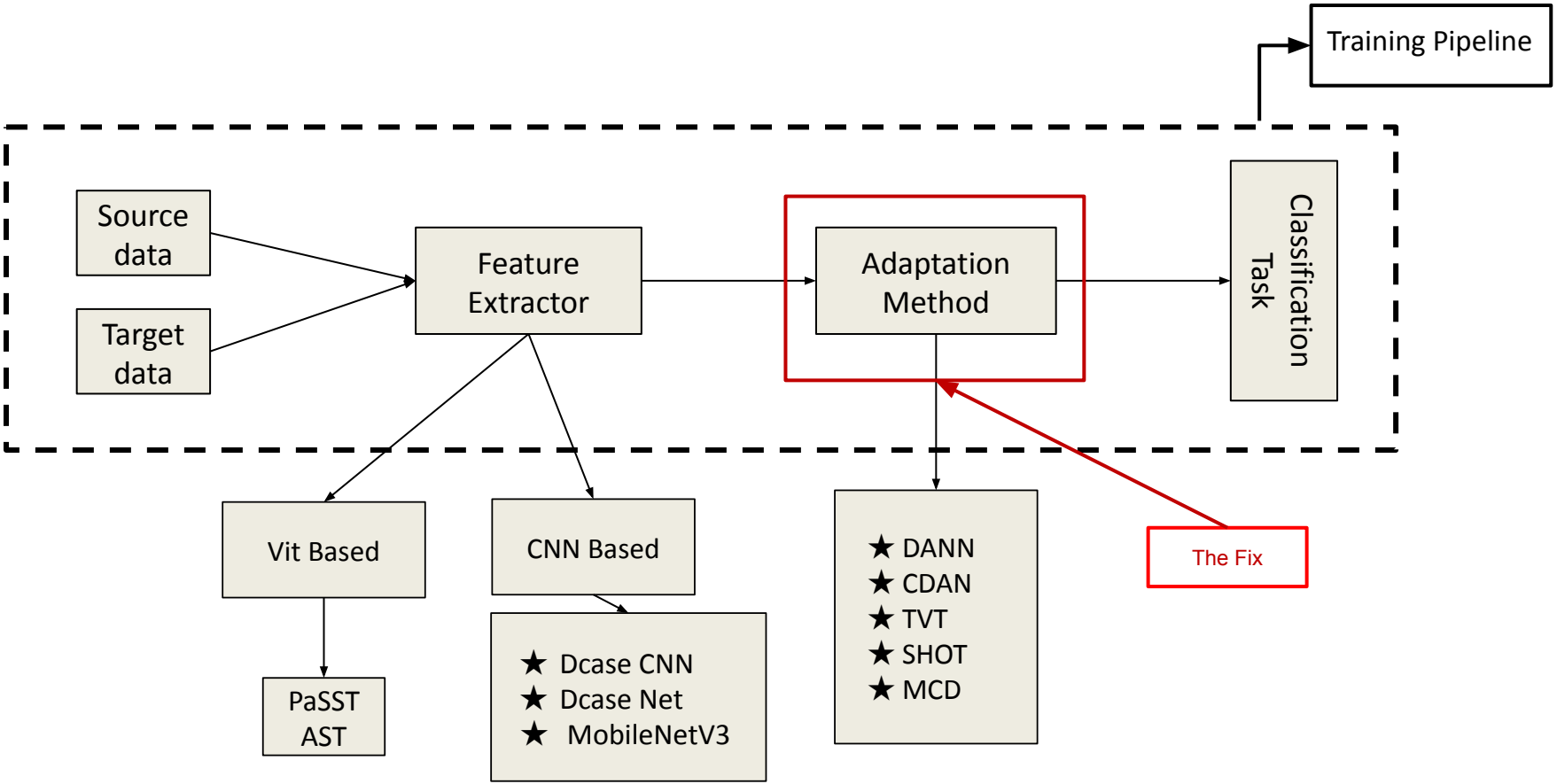
- Device A : Soundman OKM II Klassik/studio A3 (Binaural Microphone)
- Device B is a Samsung Galaxy S7
- Device C is iPhone SE
- Device s1-s3 (simulated device from these original devices)

| Source Device | Target Device | Base Acc |
|---|---|---|
| DevA | | 0.845 |
| DevA -> | DevB | 0.635 |
| DevA -> | DevC | 0.65 |
| DevA -> | DevS1 | 0.645 |
| DevA -> | DevS2 | 0.61 |
| DevA -> | DevS3 | 0.655 |



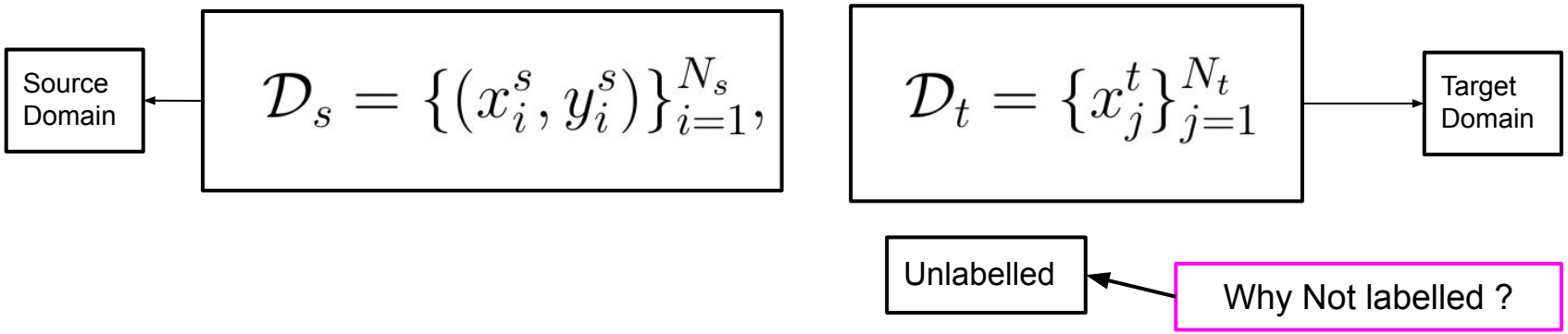
Accuracy drop

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# Problem Formulation

Let $\mathscr{D}\square$ and $\mathscr{D}\square$ denote the source and target domains, respectively




- Target domain $\mathscr{D}\square$ contains no labels $y_i$ (Unsupervised Domain Adaptation)

# Domain Adaptation Methods

**Adversarial Methods**

- Domain Adversarial Neural Networks (DANN)
- Maximum Classifier Discrepancy (MCD)
- Conditional Domain Adversarial Networks (CDAN)

**Source-Free Domain Adaptation**

- Source Hypothesis Transfer (SHOT)

**Transformer-Based Adaptation**

- Transferable Vision Transformers (TVT)

**Proposed Methodology**

- Multi-Target Adaptation via Knowledge Distillation (MTDA-KD)

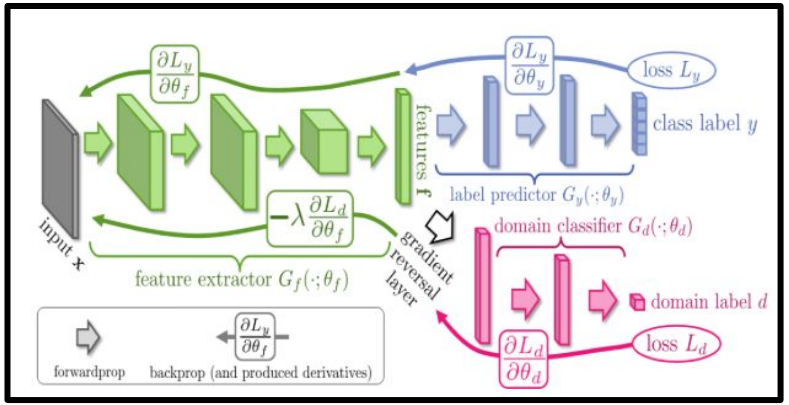
# Domain-Adversarial Neural Networks (DANN)

The overall objective function is written as:

$$\min_{G_f, G_y} \max_{G_d} \mathcal{L}_y(G_f, G_y) - \lambda \mathcal{L}_d(G_f, G_d)$$

Classifier loss

Discriminator loss




- $G_f$: Feature extractor
- $G_y$: Task classifier
- $G_d$: Domain discriminator
- λ: Trade-off hyperparameter
- Uses Gradient Reversal Layer (GRL) during backpropagation




Ganin et al., “Domain-Adversarial Training of Neural Networks”, JMLR 2016.

# Results on Vision

| Dataset | Before DA | After DA |
|---|---|---|
| SVHN→MNIST | 52.25 | 57.2 |
| Office-31 | 70.1 | 82.2 |
| Office-Home | 34.3 | 57.6 |
| VisDA-2017 | 52.4 | 57.4 |

Target accuracy (%) as reported in CDAN & MCD paper

# Results on Audio

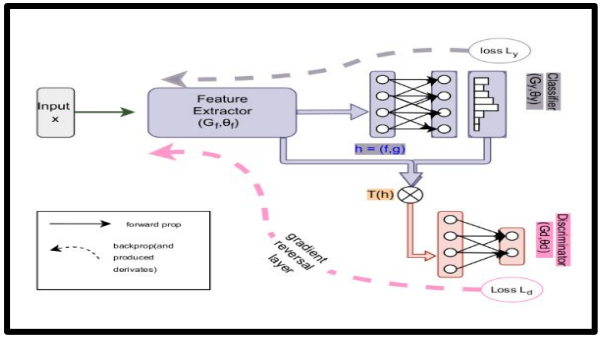
| Backbone | Dataset | Before DA | After DA |
|---|---|---|---|
| PaSST | TAU 2020 | 62.76 | 74.9 |
| PaSST | TAU 2022 | 37.9 | 53.1 |
| MobileNetV3 | TAU 2020 | 49.28 | 63.4 |
| MobileNetV3 | TAU 2022 | 35.0 | 46.6 |
| DcaseNet* | TAU 2020 | 56.7 | 69.7 |
| Custom CNN* | TAU 2020 | 24.5 | 31.5 |

Avg. target accuracy (%) over unseen devices B, C, S1–S3. * ICEI 2025; other rows ICASSP 2027.

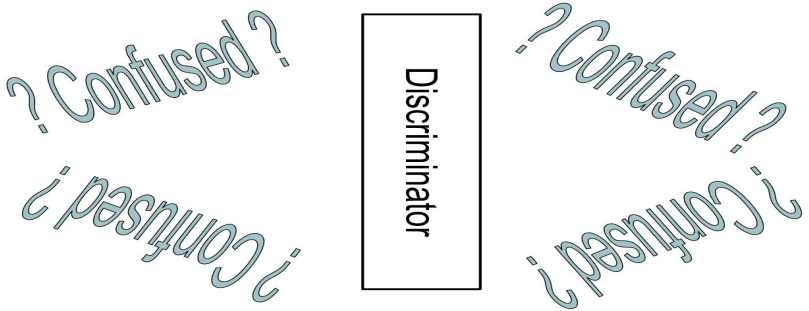
# Conditional Domain Adversarial Networks (CDAN)

The domain discriminator input is a multilinear outer product of feature representation f and classifier output g:

$$h = (f, g)$$
$$T(h) = T(f \otimes g)$$




Conditions the domain alignment on class predictions to achieve discriminative adaptation.




M. Long et al., “Conditional Adversarial Domain Adaptation”, NeurIPS 2018.

# Results on Vision

| Dataset | Before DA | After DA |
|---|---|---|
| Office-31 | 70.1 | 87.7 |
| Office-Home | 34.3 | 65.8 |
| VisDA-2017 | 52.4 | 73.9 |

Target accuracy (%) as reported in the original papers (CDAN+E).

# Results on Audio

| Backbone | Dataset | Before DA | After DA |
|---|---|---|---|
| PaSST* | TAU 2020 | 63.9 | Did not converge |
| DcaseNet* | TAU 2020 | 56.7 | 70.4 |
| Custom CNN* | TAU 2020 | 24.5 | 38.2 |

Avg. target accuracy (%) over target domain B, C, S1–S3. * ICEI 2025 .

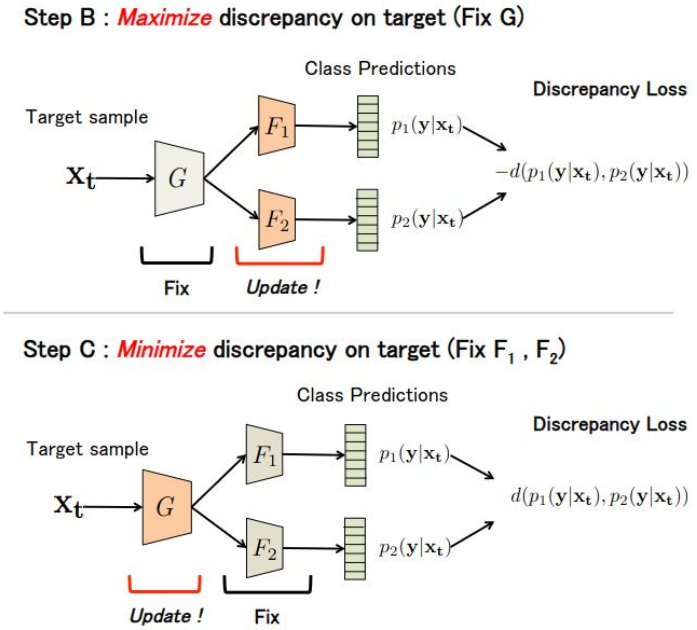
# Maximum Classifier Discrepancy (MCD)

Uses two classifiers ($G_1$, $G_2$) to detect target samples outside source support.

The adversarial loss optimizes discrepancy:

$$d(p_1, p_2) = \frac{1}{K} \sum_{k=1}^{K} |p_{1k} - p_{2k}|$$

Min-max optimization occurs in a stage-wise training procedure rather than using GRL layers.




K. Saito et al., “Maximum Classifier Discrepancy for Unsupervised Domain Adaptation”, CVPR 2019.

# Results on Vision

| Dataset | Before DA | After DA |
|---|---|---|
| SVHN→MNIST | 67.1 | 96.2 |
| VisDA-2017 | 52.4 | 71.9 |

Target accuracy (%) as reported by Saito et al. (CVPR 2019); segmentation in mIoU.

# Results on Audio

| Backbone | Dataset | Before DA | After DA |
|---|---|---|---|
| PaSST | TAU 2020 | 62.76 | 73.2 |
| PaSST | TAU 2022 | 37.9 | 50.9 |
| MobileNetV3 | TAU 2020 | 49.28 | 62.5 |
| MobileNetV3 | TAU 2022 | 35.0 | 45.7 |

Avg. target accuracy (%) over unseen devices B, C, S1–S3 (ICASSP 2027).

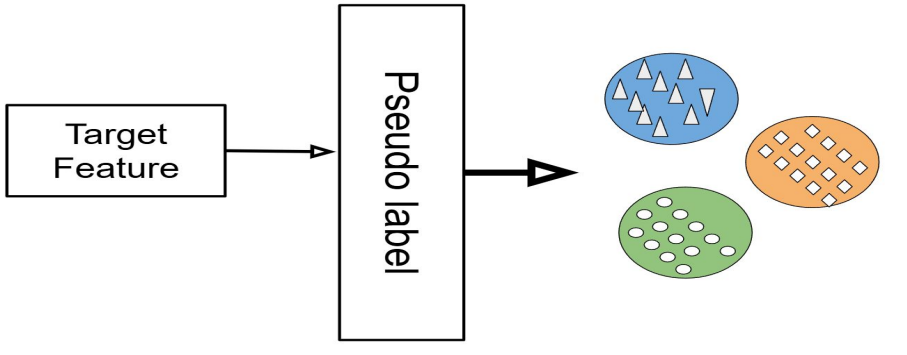
# Source Hypothesis Transfer (SHOT)

Adapts to the target domain without accessing source data during adaptation

- Freezes source-trained classifier hypothesis $G_y$ and updates feature extractor $G_f$
- Employs Information Maximization ($\mathcal{L}_{im}$) and Pseudo-Labelling ($\mathcal{L}_{pl}$)

$$\mathcal{L}_{obj} = \mathcal{L}_{im} + \mu\mathcal{L}_{pl}$$

- **Information Maximization ($\mathcal{L}_{im}$):** Combines entropy minimization (high prediction certainty) with diversity maximization across target outputs
- **Pseudo-Labelling ($\mathcal{L}_{pl}$):** Generates target pseudo-labels via class-wise prototype k-means clustering




J. Liang et al., “Do We Really Need to Access the Source Data? Source Hypothesis Transfer for Unsupervised Domain Adaptation”, ICML 2020.

# Results on Vision

| Dataset | Before DA | Accuracy (%) |
|---|---|---|
| Office-31 | 70.1 | 88.6 |
| Office-Home | 34.3 | 71.8 |
| VisDA-2017 | 52.4 | 82.9 |

Target accuracy (%) as reported in the SHOT paper (ResNet-50/101).

# Results on Audio

| Backbone | Dataset | Before DA | After DA |
|---|---|---|---|
| PaSST | TAU 2020 | 62.76 | 67.7 |
| PaSST | TAU 2022 | 37.9 | 42.1 |
| MobileNetV3 | TAU 2020 | 49.28 | 59.3 |
| MobileNetV3 | TAU 2022 | 35.0 | 45.0 |

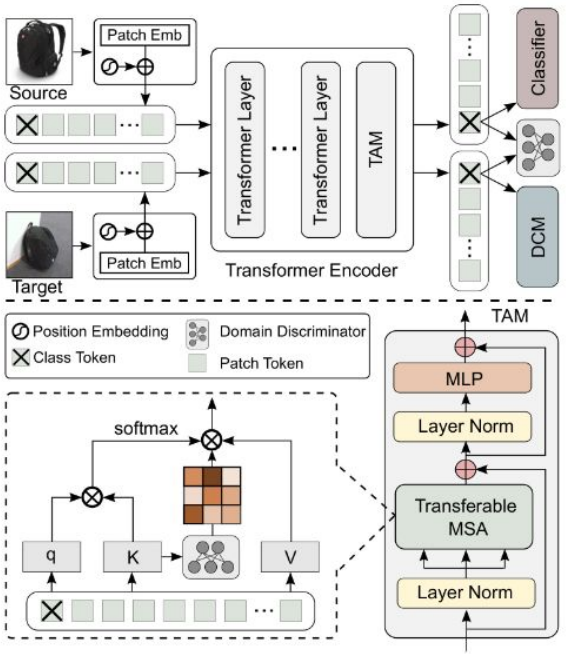
Avg. target accuracy (%) over unseen devices B, C, S1–S3 (ICASSP 2027).

# Transferable Vision Transformer (TVT)

Specifically designed for patch-based transformer architectures (e.g., PaSST)

$$\min_{G_f, G_y} \mathcal{L}_{cls}(G_y(G_f(x_s)), y_s) + \lambda \max_{D_g} \mathcal{L}_{global} + \beta \max_{D_l} \mathcal{L}_{local}$$

- Global Discriminator ($D_g$)
- Local Discriminator ($D_l$) on individual patches
- Transferability Adaptation Module (TAM) to score patches




J. Yang et al., “TVT: Transferable Vision Transformer for Unsupervised Domain Adaptation”, WACV 2023.

# Results on Vision

| Dataset | Before DA | Accuracy (%) |
|---|---|---|
| Office-31 | 70.1 | 93.8 |
| Office-Home | 34.3 | 83.6 |
| VisDA-2017 | 52.4 | 83.9 |

Target accuracy (%) as reported in the TVT paper TVT uses a ViT-B backbone pretrained on ImageNet-21K.

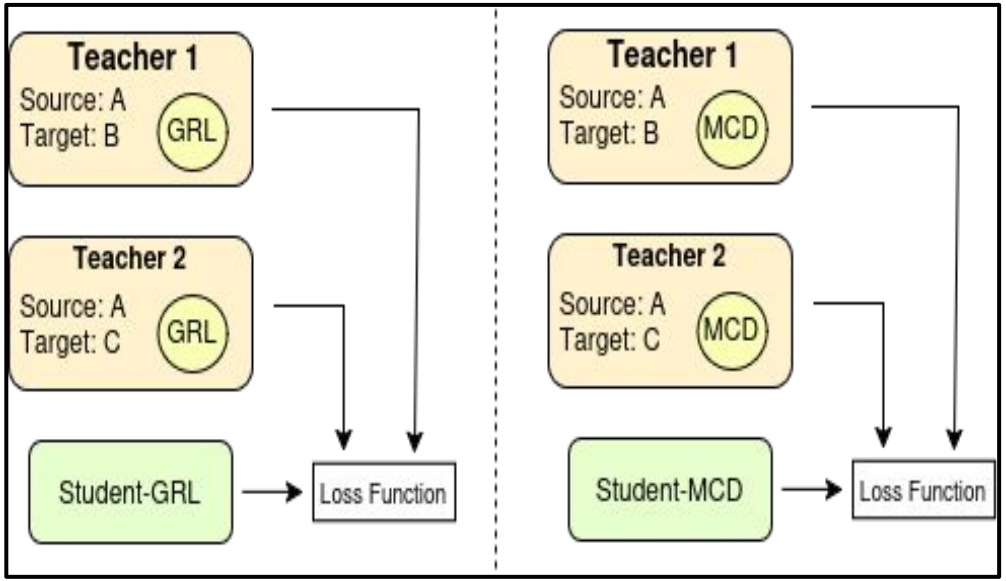
# Results on Audio

| Backbone | Dataset | Before DA | After DA |
|---|---|---|---|
| PaSST | TAU 2020 | 62.76 | 73.0 |
| PaSST | TAU 2022 | 37.9 | 48.5 |

Avg. target accuracy (%) over unseen devices B, C, S1–S3 (ICASSP 2027). TVT applies to transformer backbones only.

# Multi-Target Domain adaptation : Knowledge distillation

We use teachers from the adaptation method
Teacher A and B are domain adapted , we use the teachers logits for distillation

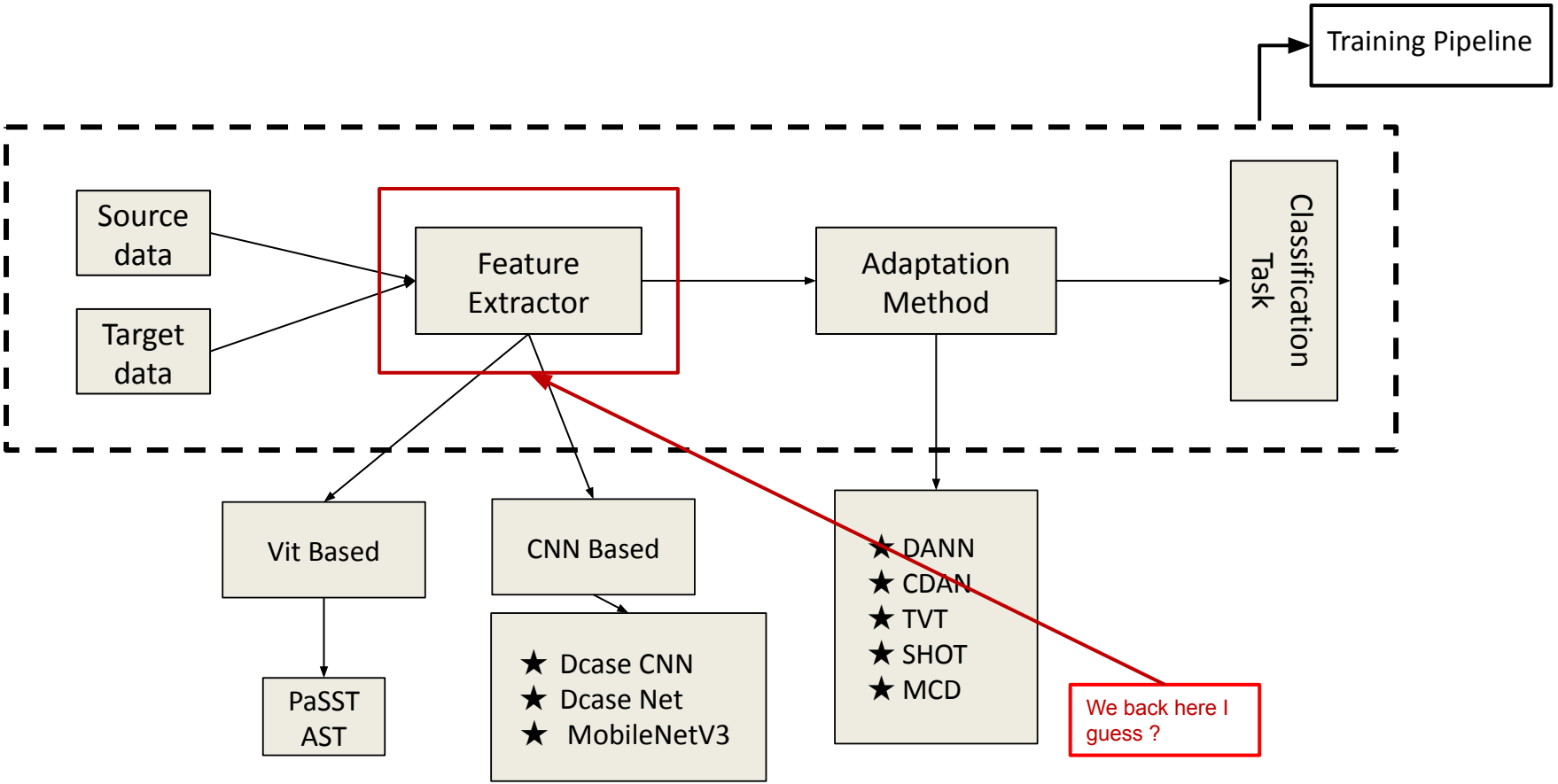


# Results on Audio

| Dataset | MobilenetV3 | A | B | C |
|---|---|---|---|---|
| TAU 2020 | Source | 83.0 | 53.5 | 72.0 |
| | Student-GRL | 82.12 | 68.39 | 77.20 |
| | Student-MCD | 83.03 | 64.13 | 72.34 |
| TAU 2022 | Source | 65.3 | 40.7 | 56.6 |
| | Student-GRL | 67.39 | 50.57 | 55.13 |
| | Student-MCD | 68.39 | 46.39 | 60.06 |

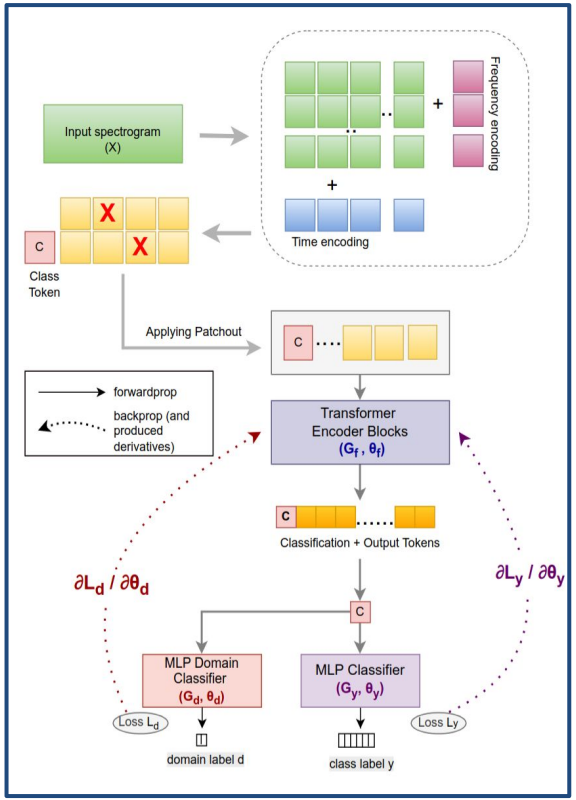
Avg. target accuracy (%) over unseen devices B, C, S1–S3 (ICASSP 2027). TVT applies to transformer backbones only.

# Overview

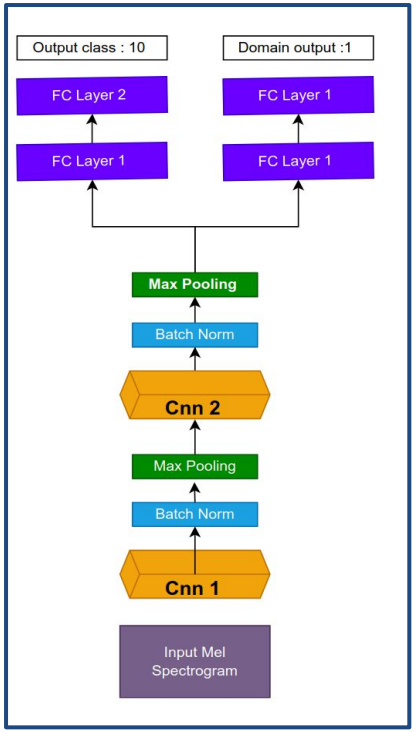

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We back here I guess ?

# Methodology : Feature Extractors

PaSST - Patch Out-Spectrogram Transformer

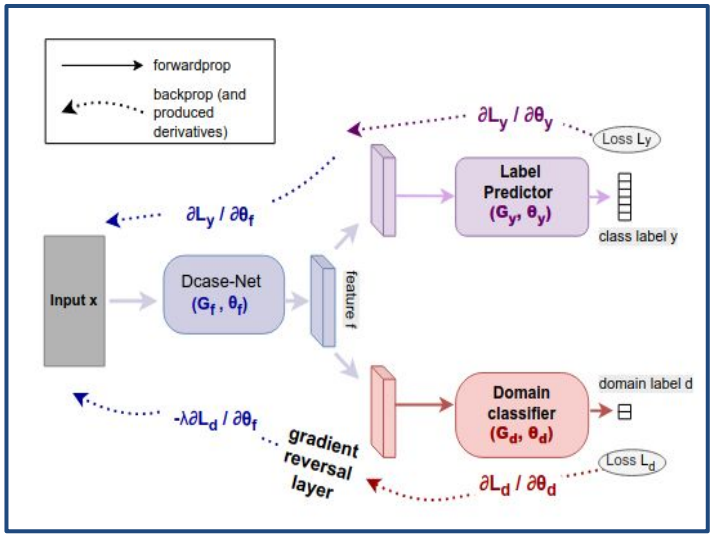


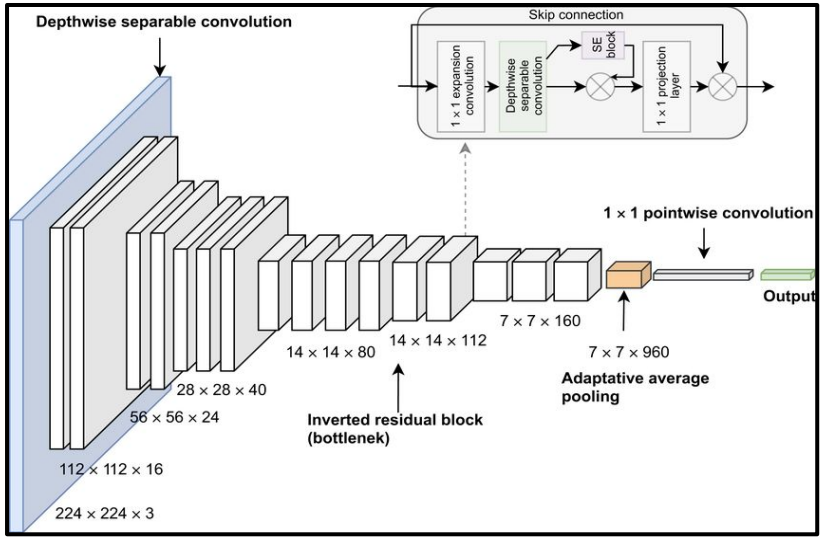

Dcase CNN Model




1.Khaled Koutini, Schlüter J, Hamid Eghbal-zadeh, Widmer G. Efficient Training of Audio Transformers with Patchout. *Interspeech 2022*. Published online September 18, 2022.

# Methodology : Feature Extractors




DcaseNet - CNN Model

Mobile Net V3


- 1.Jung J, et al. DCASE-NET... ICASSP 2021.
- 2.Elsayed Abd Elaziz, et al. Evolution toward intelligent communications... Wiley, 2023.
- 3.Schmid, Florian, Khaled Koutini, and Gerhard Widmer. "Efficient large-scale audio tagging via transformer-to-cnn knowledge distillation." ICASSP, 2023.

# Experimental Setup

- **Datasets:** TAU Urban Acoustic Scenes 2020 (10 s clips, 10 classes) & TAU 2022 (1 s clips)
- **Domains:** Source domain is Device A (40 h audio)
  - Real target devices: B, C (3 h audio each)
  - Simulated target devices: S1, S2, S3 (3 h audio each, convolved impulse responses)
- **Feature Extractors / Backbones:**
  - *Transformers:* Patchout Spectrogram Transformer (PaSST)
  - *CNNs:* MobileNetV3, DcaseNet, and Custom 2-layer CNN
- **Evaluation Metric:** Target-domain multi-class classification accuracy (%)

# Results

| Source Device | Target Device | Base Acc | GRL |
|---|---|---|---|
| DevA | | 0.845 | - |
| DevA -> | DevB | 0.635 | 0.73 |
| DevA -> | DevC | 0.65 | 0.74 |
| DevA -> | DevS1 | 0.645 | 0.65 |
| DevA -> | DevS2 | 0.61 | 0.74 |
| DevA -> | DevS3 | 0.655 | 0.74 |

PaSST Results

After Adaptation

| source device | tgt device | accuracy | GRL | CDAN |
|---|---|---|---|---|
| DevA | | 61.21% | | |
| DevA -> | DevB | 24.32% | 38.60% | 33.13% |
| DevA -> | DevC | 24.62% | 55.92% | 48.32% |
| DevA -> | DevS1 | 24.92% | 20.30% | 48.32% |
| DevA -> | DevS2 | 23.71% | 18.48% | 26.67% |
| DevA -> | DevS3 | 25.23% | 26.36% | 26.67% |

After Adaptation

Dcase CNN Results

A Dileep et al., “Device Invariance using Domain Adaptation on Acoustic Scene Classification”, IEEE ICEI 2025 (arXiv:2607.25887).

# Domain Adaptation Results (TAU 2020 & 2022)

| Method | TAU 2020 (10 s) | | | | | | | TAU 2022 (1 s) | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | A→A | A→B | A→C | A→S1 | A→S2 | A→S3 | Avg | A→A | A→B | A→C | A→S1 | A→S2 | A→S3 | Avg |
| PaSST source | 82.5 | 66.0 | 70.2 | 60.0 | 56.6 | 61.0 | 62.76 | 70.45 | 47.9 | 60.27 | 27.7 | 23.03 | 30.6 | 37.9 |
| + DIR | – | 72.0 | 69.3 | 69.1 | 62.7 | 67.6 | 68.1 | – | 52.1 | 60.7 | 42.2 | 41.9 | 44.8 | 48.3 |
| + DANN | – | 73.0 | 79.0 | **74.8** | **73.9** | **74.0** | **74.9** | – | 54.9 | 64.4 | 47.0 | 45.8 | 53.3 | 53.1 |
| + MCD | – | 71.1 | 78.0 | 73.9 | 68.8 | 74.2 | 73.2 | – | 52.7 | 63.4 | 47.5 | 42.8 | 48.3 | 50.9 |
| + SHOT | – | 72.3 | 79.6 | 60.9 | 57.0 | 68.8 | 67.7 | – | 48.4 | 57.0 | 32.4 | 30.8 | 42.0 | 42.1 |
| + TVT | – | **74.8** | **82.1** | 70.0 | 66.1 | 72.1 | 73.0 | – | 51.9 | 61.3 | 44.9 | 37.8 | 47.7 | 48.5 |
| MobileNetV3 source | 83.0 | 53.5 | 72.0 | 40.9 | 37.6 | 42.4 | 49.28 | 65.3 | 40.7 | 56.6 | 25.7 | 25.3 | 26.6 | 35.0 |
| + DIR | – | 59.5 | 71.12 | 58.78 | 55.75 | 60.3 | 61.1 | – | 40.5 | 53.3 | 37.1 | 36.1 | 39.21 | 41.25 |
| + DANN | – | **65.1** | 75.7 | 54.6 | **58.8** | **62.7** | **63.4** | – | **48.2** | **59.6** | **40.8** | 39.0 | 45.6 | **46.6** |
| + MCD | – | 62.9 | **77.2** | **56.1** | 55.1 | 61.2 | 62.5 | – | 48.8 | 58.8 | 38.4 | **40.4** | 42.4 | 45.7 |
| + SHOT | – | 59.7 | 72.9 | 52.7 | 50.0 | 60.9 | 59.3 | – | 46.3 | 56.2 | 37.8 | 37.5 | **47.0** | 45.0 |

A. Dileep, et al., “MIC DROP: Bridging Device Gaps in Acoustic Scene Classification via Unsupervised Domain Adaptation”, submitted to IEEE ICASSP 2027.

# Thank You

Questions?